SAINT MICHAEL COLLEGE OF CARAGA

FROM LAB COATS TO COURTROOM: CHALLENGES OF FORENSIC EXPERTS DURING COURT PRESENTATION

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CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the Study

Forensic experts have always been crucial to the justice system because they can simplify complex evidence in criminal cases. Forensic specialists have been integral to the legal system for a long time, as they are responsible for interpreting and analyzing complex evidence in criminal cases. However, even though they are aware of what is going on, they often struggle to remain completely objective and fair. They must maintain their objectivity while balancing the diverse needs of legal teams. They also have to explain complex scientific discoveries in a way that non-scientists can understand. This is a significant task because a mistake or bias in their evidence could alter the trial's outcome and lead to unfair results. Forensic professionals must be able to communicate clearly and accurately to promote justice and foster public trust in the legal system (Weyermann & Roux, 2021; Melnyk et al., 2022). Many prosecution cases depend on scientific and technical Analysis, so the experience and quality of forensic services can also affect the trial's outcome (Lytvyn et al., 2022).

Additionally, forensic science results are typically presented in written reports. If the case goes to trial, forensic scientists may be asked to provide testimony based on the information in the report (G. Edmond et al., 2017). Several problems with forensic science have been highlighted in publications read worldwide. These problems include the way that experts report their findings and then testify about them. It is difficult to

determine the exact number of wrongly convicted individuals who have been helped by forensic science communication. However, a significant study conducted in the US ten years ago found that forensic scientific testimony was used in 156 cases where people were later acquitted.

Additionally, researchers found 137 transcripts from the trial. Of these, 60% had testimony that was deemed flawed or invalid because it included incorrect frequencies or statistics, the use of statistics or verbal expressions without empirical support, and conclusions that were not backed up by evidence (for example, that the evidence came from the defendant). Also, in England and Wales over the past 7 years, 32% of the 218 cases that were successfully appealed involved forensic evidence (N.M. Smit et al., 2018).

Researchers in Australia have identified 71 cases of individuals who were wrongfully convicted, and in 22% of those cases, false forensic evidence was used (R. Dioso-Villa, 2015). As the research subject, the Study fits into the new field of forensic criminology, which examines how forensic science is applied in society. The article begins by examining how forensic expertise is socially constructed and utilized within the criminal justice system. It then examines studies that have already been conducted on expert reports and testimony. It then discusses how forensic biology was utilized in 10 major crimes against individuals that were heard in the Tasmanian Supreme Court (A. Williams, 2015).

The Study also looked at expert reports and testimony from these specific major crime cases. Its goal was to determine if the expert reports and testimony contained the types of information that courts, forensic standards, and research indicate should be included. The Study examines how forensic science is currently practiced, with a focus on enhancing its representation in both professional settings and academic research (Reid & Howes, 2020).

1.2 Theoretical Framework

The Decisional Dimensions Theory and the Communication

As the study "From Lab Coats to Courtroom: Challenges of Forensic Experts During Court Presentation" demonstrates, Accommodation Theory can help us understand the issues forensic experts face when they must present evidence in court. All of these theories help make sense of the complicated connection between the scientific duties of forensic experts and the legal setting in which they give their testimony.

The Decisional Dimensions Theory, developed by Biedermann, Bozza, Taroni, and Garbolino (2018), examines the role of expert testimony in the legal system in decision-making processes. When making decisions, jurors and judges must consider and apply the scientific evidence provided by forensic experts. The theoretical framework emphasizes that presenting expert evidence involves more than simply stating the facts; it also entails explaining probabilistic data and clarifying the uncertainty surrounding scientific findings. Because of this, experts must carefully balance upholding scientific truth with meeting the

legal system's needs for accuracy and usefulness. This disagreement highlights the challenges faced by juries and judges in comprehending forensic evidence. It also demonstrates that experts have a moral obligation to present their findings in a manner that facilitates fair and well-informed legal decisions.

The Communication Accommodation Theory, first proposed by Giles and Gallois and updated in 2023, also explains how people adjust their communication to fit the needs of the people they are interacting with. When they are in court, forensic experts collaborate with judges, jurors, and lawyers who may not be well-versed in scientific principles.

In this situation, communication only works if the experts adjust their approach to presenting information so that it is easy to understand the complex ideas without losing the point of the conversation or coming across as too pushy. Depending on the court case, the theory suggests that experts either adapt their communication style to convey credibility and establish rapport, or they may opt for a professional tone to remain clear and support their case.

Understanding these theories helps forensic professionals better comprehend the challenges that arise in court, highlighting the difficulties they face in balancing their scientific work with effective communication within the court system. The theoretical basis strengthens the push for better expert testimony, ultimately making legal decisions fairer and more accurate in the long run.

1.3 Review of Related Literature

Forensic criminology, also known as forensic studies, aims to determine what happened between the crime scene and the courtroom, enabling a better understanding of how the criminal justice system operates (A. Williams, 2015). This field of Study has been sparked by past injustices and ideas from the sociology of science (C. Lawless, 2016). As D.S. Caudill (2018) points out, most forensic fields were not established by scientists but by the criminal justice system to aid in investigations.

Evidence is only gathered when crimes occur and court cases are ongoing. It is not available before the fact, like research evidence often is (A. Williams, 2015). Aside from DNA, forensic scientists have not had to demonstrate reliability through study replication, unlike researchers in other fields (D.S. Caudill, 2018). Forensic science can make mistakes, but it may never reach the point where it can be scrutinized by everyone and corrected like research science does (D.S. Caudill, 2018). Even with these problems, forensic science is still used along with other types of evidence.

The literature indicates that scientists in the witness box must navigate the tension between being precise and being easily understood. This is because the court always wants expert testimony to be clear. Koch (2025) highlights the primary challenge: explaining complex forensic findings in a manner that a layperson, often under considerable stress, can comprehend and evaluate. When experts think their audience

knows more about their method and terminology than they actually do, they hurt trust and, ironically, put at risk the evidence-based value of otherwise sound science.

Reid and Howes (2020) further illustrate this point by stating that reports that exceed the jury's ability to analyze are met with skepticism. In contrast, reports that omit important details lack supporting evidence. If the experimental process and the story are not aligned correctly, it leads to more perception problems and hurts the credibility of all decision-makers. The 2019 Judicature survey supports this by citing case records that show judges, litigants, and jurors have different levels of scientific literacy. After the fact, testimony that was accepted in one courtroom is rejected in another, highlighting the repeated and systemic consequences of experts who are either unprepared or overly prepared. This illustrates the challenges courts face when evaluating forensic evidence that does not consistently meet rigorous scientific validity standards. Because of this, people who are suing must demand that experts explain their conclusions in clear language and ensure that their testimony adheres to the rules outlined in the Rules of Evidence. Increasingly, people are highlighting that forensic reports are a unique type of document that must adhere to both the law and the principles of the scientific method (Higley et al. 2025).

The Crown Prosecution Service (2023) says that the clarity of expert evidence is essential. If the evidence is not clearly stated and put into context, it is more likely to be misunderstood, especially when decisions are based on new forensic evidence. When this

happens, a muddy presentation can make people think they know for sure when they do not. Baxter et al. (2025) support unconventional ways of disseminating information and suggest creating short comics that turn forensic Analysis into storylines. The information is presented visually, broken down into frames that are easy to understand, while maintaining the core ideas and adapting the language for a jury-level audience. This creative shift, they say, could change how juries think about things from abstract math to progressive stories, lowering the bar for understanding and remembering (O'Brien et al., 2015), which is less positive about radical media but shows how weak crystalline testimony is: robust evidence is often replaced with useful shortcuts that cut down on uncertainty.

The authors suggest that pedagogical inoculation, in which forensic professionals regularly practice and critique the story in a simple courtroom style, evolves into a safety measure rather than a style indulgence. According to O'Brien et al. (2015), the only way to mitigate the threat of misapplied science is for clinical interpretation to become clearer over time. This training should strike a good balance between scientific rigor and the ability to explain forensic concepts to non-experts (O'Brien et al. 2015).

Because it is hard for forensic experts to translate scientific results for the courtroom, everyone needs to work together. Enhancing the understanding and reliability of forensic evidence in court can be achieved by employing effective messaging techniques, innovative methods of presenting evidence, and a commitment to ongoing training. Since

the field is constantly evolving, professionals must adapt their approach to communicating with the public and the courts. This ensures that scientific data is presented in a transparent and trustworthy manner, which is essential for obtaining a fair verdict. Marquis et al. (2016) state that all necessary information to understand should be presented clearly and without any doubt. For example, a chart of possible prior probabilities, the LR presented in the case, and the updated posterior probabilities should all be shown.

Not using any statistics and instead using other methods, some of which she lists, to show how strong the evidence is. Again, these strategies should be tested on potential jurors to assess their effectiveness. 5) Each Study was mostly about a single or a few types of evidence or crimes. There are clear reasons for this, but research indicates that jurors interpret evidence differently, depending on the type of evidence and possibly even the type of crime, due to their existing beliefs and values. It is essential to conduct comparative studies to identify which effects are consistent across all crime/evidence combinations and which ones may vary, as well as to understand how they differ (Spellman, B. A., 2017).

Expertise and Credibility

Forensic expert credibility is multifaceted, established through qualifications, experience, communication skills, professionalism, ethical behavior, and transparency. Studies emphasize advanced education, specialized training, certifications, and years of

experience as key factors enhancing credibility. Jurors often associate age and career length with expertise, but also recognize younger experts' advantages, such as familiarity with cutting-edge technology. Trustworthiness during testimony is influenced by demeanor, confidence, ethical transparency, and adherence to forensic standards, highlighting credibility as a dynamic combination of multiple factors, not just time served (Koodrin, 2020).

In Addition, what is also essential for forensic experts in the context of the discussed works is their role in comprehending the essence of the cases under consideration when presenting the evidence in a form readily understandable for a layperson (D'Orio et al., 2022). This role is vital, especially in trials involving technical evidence, such as DNA, Toxicology, or digital evidence (D'Orio et al., 2022; Ishihara & Iwase, 2020). The literature points out that they focus on forensic testimony, stating that the accuracy and clarity of this evidence are critical in these cases since any misinterpretation of the evidence could lead to problematic judicial decisions. Forensics is becoming more advanced as days pass, and these professionals greatly help ensure that justice prevails through the laboratories.

However, the literature also depicts several issues emerging with bias in forensic testimonies. Various scholars have discussed this aspect, and one of them is Jones and White (2020), who opined that forensic officers, like any other person, cannot escape biases that may affect their work. Such biases can originate from the expert's perception,

the client or the institution that hired them, or from the very nature of a legal battle. For instance, in adversarial legal systems, the expert might be employed to defend a particular side and hence may unwittingly or willingly aid the side that hired him in presenting their evidence in a certain way. This potential for bias has provoked discussions on increasing the training, certification, and standardization of forensic methods to ensure that the specialists have no self-interest in their findings.

Williams et al. (2022) and Rodriguez and Singh (2023) also mention the problems of a forensic specialist: to be aware of innovations and adhere to strict procedures. Green and Martin (2021) point out that variations in the quality and practice between one country and another may affect the accuracy and efficiency of forensic evidence. The document analysis of the study also validates some similarities. However, there are differences in forensic procedures, for instance, concerning qualifications and where skills and knowledge need to be updated. Technological advances and proficiency standards that forensic experts in the United States test against present moderate levels of difficulty for these professionals, moderate to high for German ones, and high levels of challenge with variable standards for Japanese experts. According to Jenkins et al. (2023), such fluctuations imply the need to develop more homogenized processes worldwide.

Due to concern about the credibility of forensic expert opinions, some researchers have argued that forensic experts should report both broad and narrow assessments of accuracy, reliability, and validity (Edmond et al., 2016; PCAST, 2016). Narrow assessments

of validity focus on the witness (juror) level and how these witnesses evaluate the credibility of an expert's opinion (Edmond et al., 2016). Some researchers have been exploring factors that affect perceptions of expert credibility at the witness level, in order to improve the assessment of forensic expert opinion (Martire et al., 2020).

Based on prior findings in studies looking at forensic expert credibility (Martire et al., 2020), ratings of expert credibility as they are affected by framed expert performance data will be measured. Comparing the ratings of experts' credibility will indicate how effective the framed data was in influencing juror decisions. Understanding how different presentations of expert performance data affect assessments of expert credibility will increase our understanding of how juror decisions can be influenced by performance data. Further, the findings from this thesis may provide empirical evidence to encourage openness of forensic performance in the courtroom (McAndrew & Houck, 2020; Searston & Chin, 2019).

Communication and Presentation

The ability of forensic experts to effectively communicate scientific findings to jurors are critical in ensuring accurate comprehension and meaningful impact in legal decisions. Research indicates that jurors, often lacking a scientific background, require expert testimony to be presented in accessible, clear language supplemented with relatable analogies and visual aids such as graphs or charts. Courtroom experience plays an important role in helping experts develop the skill to adjust their explanations to the

lay audience, making complex evidence understandable without compromising technical accuracy (Eldridge, 2019).

Older experts typically possess more courtroom exposure, allowing them to refine their communication style and develop strategies to engage jurors effectively. Younger experts, while sometimes less polished in courtroom demeanor, may compensate by using modern technology-enhanced presentations and thorough trial preparation. Integrating multimedia tools and clear, structured narratives can significantly boost juror comprehension and, consequently, the persuasive power of the testimony. Emphasizing the importance of ongoing training and preparation in enhancing communication skills and courtroom effectiveness for forensic experts. Professionals who continuously work to improve their public speaking and explanation methods tend to foster greater credibility and influence among juries (Eldridge, 2019).

Communication of forensic expert evidence has tended to focus on how experts present their source conclusions to the courtroom, noting that the source conclusion is the final decision an expert presents to the jury. For example, DNA experts may use Random Match Probabilities (RMPs) to express the likelihood that an unrelated person, randomly sampled from the general population, would match the DNA sample found at the crime scene (Eldridge, 2019; Thompson, Grady, Lai, & Stern, 2018). Firearm and fingerprint experts, on the other hand, tend to make categorical statements about the source of trace evidence (Thompson et al., 2018).

For example, a fingerprint examiner will conclude a fingerprint as being identified or individuated as a particular person or possession of a person (Thompson et al., 2018). Recently, researchers have begun questioning the accuracy of the science behind forensic experts' methods to testify within the courtroom (e.g., the likelihood that the defendant's fingerprint matches prints left at the crime scene (Thompson, 2018).

Expert Reports

Research reveals differences in the types of information that various US jurisdictions, laboratories, and scientific fields include in their expert reports. Some expert reports have been called "certificates of analysis" because they leave out methodological details, conclusions, and the Analysis's limitations. There is also a lack of important information in expert reports in Australia. Reports had problems with omitting details about the assumptions and limitations of scientific methods when discussing topics such as DNA analysis and forensic examination of glass. These empirical studies support Recommendation 2 of the NAS Report, which states that court reports should have more consistent content and that the language used to express opinions should be standardized (using templates). This would make it more straightforward to understand how uncertain the results are. Standards Australia released a document in 2013 titled "Forensic Analysis: Part 4 – Reporting," which explained the contents of expert reports. Practice notes, such as those made by the Supreme Court of Tasmania (ANZPAA NIFS (2017) and the Federal Court of Australia, also explain what should be in the report.

The Australian and New Zealand Policing Advisory Agency and the National Institute of Forensic Science have released an introduction to evaluative reporting, which discusses how to present your opinion while still acknowledging uncertainty. There is a global trend (see European Network of Forensic Science Institutes (ENFSI), 2015) for more reports to include likelihood ratios. These are numbers that indicate the likelihood that one hypothesis is more accurate than another. Additionally, researchers have provided detailed suggestions on what should be included in expert reports. There is scientific and practical information, including details about the forensic lab, a list of evidence, information about the chain of custody, descriptions of the method and its limitations, descriptions of the data collected, and steps for collecting and analyzing the data (G. Edmond et al., 2016).

There have also been suggestions to help people understand the report better by using words that adequately show uncertainty (for example, not reporting in absolute terms like "100% certainty" as was talked about in The Fingerprint Inquiry), letting people know about any relevant scientific controversy, outlining limitations and possible sources of error, and mentioning steps taken to reduce biases (G. Edmond et al., 2016).

To make things easier for the reader, the report should include a purpose statement, instructions on how to use the report, a complete summary, enough information about each step of the process presented in a logical order (L.M. Howes, 2015), and a description of the whole scale used to show the strength of the evidence.

Most suggestions say that expert reports should have more information (G. Edmond et al., 2016). It is essential to explain scientific concepts in a manner that is accessible to the general public. It is essential to provide detailed explanations and links to other resources, as people who are not experts often lack the specialized background knowledge you possess. Scientific writing tends to be concise, but reports also need to be transparent about the limitations of their methods, techniques, and interpretations. People who are not experts in the field may obtain incorrect results or mistakenly believe them to be more reliable if they are not clearly presented. These things may not be known to regular people, such as lawyers, so that they may be brought up in court (G. Edmond et al., 2016).

Researchers have identified several reasons why police and lawyers may not contact forensic scientists for additional information, even if they do not understand the expert's report or the accompanying statistics. For police officers, these reasons include not knowing who to contact and the concern that scientists would be distracted from their work if they did. For lawyers, these reasons include a lack of time and resources. Police investigators and lawyers may not ask for more detailed reports, even if they are available. This could impact how they utilize the reports to inform decisions during investigations and when presenting the reports in court (Reid & Howes, 2020).

Notably, the New South Wales (NSW) Police Force revised its expert reports following the defense's recent questioning of whether fingerprint evidence could be used in court. Having more information in the expert testimony than in the report (G. Edmond

et al., 2017) is not helpful in the early stages of the criminal justice process, like the police investigation. Details should not be saved until the expert testifies in court. In the end, the fingerprint revisions expanded from a 3-page report to an 8-page report, incorporating significantly more information that aligns better with the Australian Standard Forensic Analysis: Part 4 – Reporting and the NSW Expert Witness Code of Conduct (G. Edmond et al., 2017).

Expert Testimony

Several suggestions have been made based on research into testimony. The most important thing is that forensic scientists should be able to explain how they followed all the court's rules of behavior when giving expert testimony (A. Blow, 2016). Similarly, scientists should not present too much evidence when expressing their opinions; instead, they should be "epistemologically modest" (G. Edmond et al., 2016). On the other hand, forensic scientists cannot give their expert opinion and explain why they think that way unless the prosecution or defense lawyer asks them the right questions. One Study asked forensic biologists to be cautious about how they presented their evidence, as the prosecution or defense could make it appear too strong or too weak (L.M. Howes, 2015).

The Study pointed out that if the prosecutors had not done this during the examination, cross-examination by the defense could have helped provide a more balanced picture of the evidence. This indicates that both forensic scientists and lawyers have a responsibility to ensure that forensic scientific evidence is presented in a manner

that is easily understood at trial (Reid & Howes, 2020). A long-standing problem in academic research has been determining the strength of expert evidence. Experts' use of certain words can unfairly convince and sway non-experts, sometimes making scientific methods sound more reliable than they really are. For instance, labeling the similarity between a sample from a suspect and a sample from a crime scene as a "match" may not be entirely clear. Some juries might think that this language makes it clear that the defendant is the source of the crime scene evidence, even if scientific. There is no way to know for sure what the Analysis shows because it only shows a degree of correspondence. Some people have doubts about the use of words like "match" and "consistent with" when comparing hair, fingerprints, and fibers (Reid & Howes, 2020).

However, the word "match" could imply that certain aspects of the samples are similar, suggesting that further testing might be worthwhile. This often means peaks that are the same at one or more of the loci (specific DNA sites) that are being compared in DNA profiling. Let us consider, for example, that there is a match at the peaks of all 21 loci analyzed, which is what some Australian labs do. That being the case, this provides strong evidence that the two samples originated from the same person or an identical twin. Still, mistakes can occur in the lab, such as incorrect labeling, contamination, or sample loss (Reid & Howes, 2020).

Errors can also happen when only small amounts of DNA are found or when complex mixtures are misinterpreted. Additionally, it is essential to note that DNA found

at a crime scene is only evidence of a potential crime, not conclusive proof of guilt. Just because it is there does not mean that the person did the crime.

Additionally, it is impossible to determine when or how the DNA was introduced. People on the jury need to know this so they do not think science is perfect and cannot be wrong. Researchers have explored how to utilize statistics to demonstrate the strength of the evidence, considering uncertainty correctly and appropriately. However, most people are not very good at using statistics to make decisions. A layperson might think that the probability that evidence will come from a particular source, which fits with a prosecution hypothesis, is the same as the probability that the hypothesis is true. The mistake of thinking that the defendant was guilty based on the statistic given, whether it was a likelihood ratio or a random-match probability (Reid & Howes, 2020).

This incorrect idea is known as the "prosecutor's fallacy." Approximately 92% of judges, 80% of lawyers, and 58% of expert witnesses fell for the prosecutor's fallacy in a study that utilized the likelihood ratio to demonstrate the value of forensic evidence in a simulated case. People who work in criminal justice may be more familiar with likelihood ratios, which are increasingly being used, but jurors still struggle to understand them (Reid & Howes, 2020).

Coping Mechanism

The process of becoming an expert in any domain (including forensic science) entails acquiring more knowledge, training, and experience (R.M. Morgan, 2017) so as to gradually develop finely-tuned top-down mechanisms that improve performance beyond that of novices (B. Growns, 2020). That is to say, experts develop heuristics (mental shortcuts, such as selective attention) that enable them to process information in a more automatic and efficient way and thus complete tasks more quickly and easily. However, the same top-down mechanisms that allow experts to process information very efficiently can also create vulnerabilities that increase the risk of misinterpretation —and in some cases, error (B. Growns, 2020).

One such vulnerability is cognitive bias, which refers to "the class of effects through which an individual's pre-existing beliefs, expectations, motives, and situational context influence the collection, perception, or interpretation of evidence, or their resulting judgments, decisions, or confidence. For example, if a forensic expert begins an analysis with an expectation of what the result will be, they may experience "tunnel vision", such that they search for information that fits their expectation and ignore or disregard information that refutes their expectation, which can lead to a misinterpretation (B.A. Spellman, 2022). Thus, top-down mechanisms are useful to make efficient decisions, but they can also contribute towards systematic errors due to human factors issues, including stress.

Expert training, mandatory special instruction and certification or peer-review of expert testimony (Bernstein DE et al., 2015) are meant to ensure high quality expert testimonies and make them more reliable. Efforts have also been made to improve the communication between scientific and legal experts, for example by standardizing written testimonies and adapting them to legal experts' needs (AAPL, 2015). The use of impartial experts appointed by the court or employed by the state as well as defining codes of conduct for experts are attempts to reduce potential expert bias. Furthermore, allowing more than one expert witness to address the court (52) helps to avoid relying upon one single, potentially flawed statement. These measures may help narrow the knowledge gap by weeding out unusable evidence and ensuring impartial expert witness statements of high quality. However, as stated above, the problem of conflicting evidence remains. If judges and juries are to avoid delegating too much responsibility to a scientific expert for a legal ruling, then it is important for legal professionals to decide in the face of diverging medical statements which one to follow(Canela et al., 2019). This arguably entails possessing a modicum of knowledge of the science in question, such as the understanding of medical terms and the procedures and limitations of scientific methods. In addition, they need to be able to critically appraise and understand scientific information, one of the definitions of scientific literacy (Canela et al., 2019). Previous studies have investigated different aspects of this topic; for example, how judges deal with expert knowledge focusing on the Daubert criteria, the characteristics and criteria that are used

by judges to evaluate expert testimony, how judges evaluate scientific knowledge (56) or behavioral factors of expert testimony (Canela et al., 2019).

CSIs also exhibited more stress when on a rotational schedule, which could be attributed to a lack of social support from consistent coworkers. Another feeling that was shared by CSIs was a feeling of being unappreciated by their organizations. These stressors were mirrored in the study involving CSIs (Sollie, H et al., 2017). Both studies indicated that CSIs had an immense amount of job satisfaction but felt that standby and rotational shifts negatively impacted their personal life, adding to burnout. When surveyed, these CSI had several resiliency strategies to help them cope with stress, which included emotionally distancing themselves, participating in decision-making at a crime scene, seeking support from colleagues and team leaders, and the use of humor (Clark, R. D et al., 2015).

The use of humor as a coping mechanism is a common theme in many studies. For example, in a qualitative study performed police officers did not feel the need to talk about negative events as it made them feel weak(Slack, 2020). Conversely, they recognized that keeping it “bottled up” was not healthy, and even stated that they would give advice to the new recruits to not bottle up their emotions. While this may seem like a contradiction, it is likely that these individuals were accustomed to venting their own bottled up feelings through the use of humor. They noted that using humor helped them cope, as long as the humor did not go too deep (Slack, 2020).

Slack, (2020)also noted that humor was a strong coping strategy with ICAC investigators, and even distinguished between light-hearted humor and gallows humor. The latter refers to humor that is neither dark nor light, but rather makes fun of disastrous situations. It is often only used around certain company because it can be seen as inappropriate, even though it is never at the expense of a victim. The researchers recruited their survey participants through a direct invitation to 61 ICAC commanders in the United States, who then sent the survey link to both sworn and civilian task force members. A total of 508 ICAC investigators responded and the majority of respondents were white (92.5%), married (79.2%), men (74.0%) with children (83.3%). Results indicated that gallows humor was used or witnessed by 97% of the investigators surveyed in this study. The research also indicated that humor as a coping mechanism reduced burnout and increased bonding with coworkers. Although gallows humor was used, lighthearted humor was used more frequently. Results also demonstrated that the use of lighthearted humor had the same coping effects as talking to a coworker. However, it was also noted that gallows humor, when overused, could serve as a "yellow flag" as it could be indicative of higher levels of STS in those individuals using it as a coping mechanism.

Slack, (2020)noted from their study, one strong coping strategy that is commonly used, aside from humor, is talking with a coworker. M.S. Birkeland, et al., 2017, surveyed 421 police officers to determine stigmas surrounding talking about one's feelings. For this study, one large municipal police force in Canada was surveyed. A total of 77% were male, and the average length of time on the force was 16.4 years. The researchers determined

that officers must have a supportive environment before they feel a willingness to talk. Also, officers did not feel comfortable talking to just anyone, including professionals. They prefer family and close friends, but will seek out a professional if those avenues are unavailable or not effective. However, police officers felt completely comfortable debriefing about a scene or an incident with peers who were also involved in the event. Therefore, one of the recommendations that describe is for organizations to make mandatory debriefings with peers, as this seems to have little to no stigma associated with it. A major theme in the literature appears to be the need for strong social support, not only from peers, but from leaders as well. In a longitudinal study performed by (Birkeland, 2017) it was discovered that the support from leaders was more important than general support from friends or family and is correlated with lower levels of psychological distress.

Other significant positive coping strategies include partaking in activities such as exercise, prayer, meditation, and mindfulness. J.B. Kaplan et al., (2017), describe the relationship between stress and dispositional mindfulness in law enforcement personnel. Dispositional mindfulness includes three activities or states that make one keenly aware of their own thoughts and feelings while they are in the moment. These three states are acting with awareness, nonjudging of the inner experience, and nonreactivity. In their study, the researchers invited 72 law enforcement personnel to an 8-week mindfulness-based resilience training course to train them on dispositional mindfulness. Participants were recruited from one police department in the Pacific Northwest and 57% were male, 81% identified as Euro-American, the average age was 43.5 years old, and the length of

time on duty was 13.3 years. They found that personnel low in nonreactivity exhibited a significant relationship between stress and their perception of stress. Findings such as these could help inform future interventions to include mindfulness training and counseling. In a recent article by A.M. Jeanguenat et al., (2018), it was noted that the use of mindfulness in a work setting could positively impact decision-making and focus in forensic science practitioners. This type of self-care is impactful for those who have begun to question the meaning of their work, do not know their purpose, and have internalized a victim's trauma. Mindfulness and engaging in one's own spirituality have both been shown to have a positive effect on those suffering from VT, or other forms of trauma (Slack, 2020).

Schematic Diagram

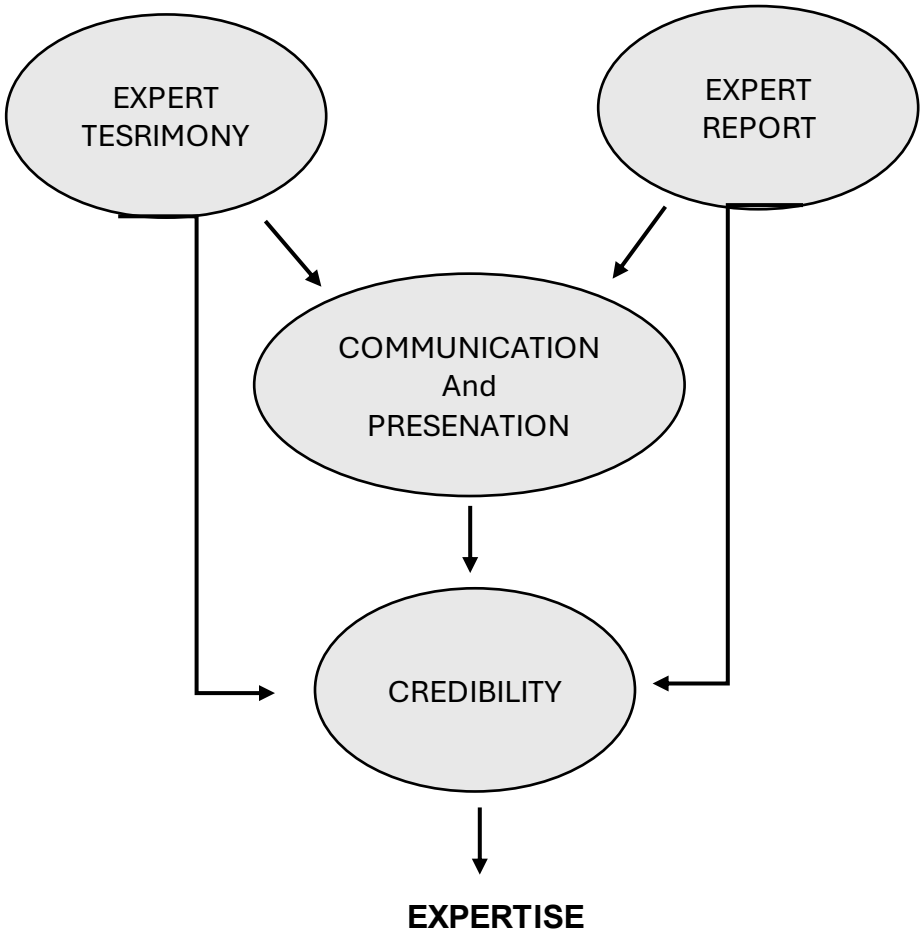


Figure 1.

It visually represents the relationships among key variables or concepts in research, illustrating how they interact and influence one another. In this case, it maps out the connections among Expert Testimony, Expert Reports, Communication and Presentation, and Expertise and Credibility.

1.4 Statement of the Problem/ Objectives of the Study

This Study aims to bridge the gap between forensic laboratory work and compelling, credible courtroom testimony, addressing the unique challenges forensic experts face in court.

Specifically, this Study sought answers to the following questions:

1. To identify the major challenges forensic experts, encounter when presenting scientific evidence in court.

2. To examine how legal procedures, courtroom dynamics, and interactions with legal actors influence the effectiveness of forensic testimony.

3. To assess the preparedness, training, and institutional support available to forensic experts for court presentation.

4. To propose strategies and recommendations for improving forensic experts' courtroom presentation skills to enhance understanding and influence trial outcomes.

1.5 Definition of Terms

Admissibility. The legal standard that determines whether forensic evidence or expert testimony is allowed in court based on its relevance and compliance with rules of evidence.

Credibility. The degree to which forensic experts and their testimony are trusted and believed by judges and jurors, based on the expert's honesty, professionalism, and accuracy.

Communication. The clear and effective explanation of complex forensic findings by experts to legal professionals and jurors to ensure understanding during trials.

Expertise. The specialized knowledge, training, and experience forensic experts possess to analyze evidence and provide authoritative opinions in legal cases.

Expert Reports. Written documents by forensic experts summarizing their analysis, methods, results, and conclusions for use in court.

Expert Testimony. The verbal explanation of forensic findings and professional opinions by experts in court to help judges and jurors comprehend technical evidence.

Evidence. Objects, documents, analyses, and testimonies presented in court to prove or disprove facts, especially scientific findings from forensic experts.

Forensic Expert. A trained scientist who applies validated methods to analyze physical evidence and provide objective findings for legal proceedings.

Forensic Science. The application of scientific methods to collect, analyze, and interpret physical evidence to support investigations and court cases.

Jurors. Community members who impartially assess evidence and expert testimony to decide case outcomes based on the facts presented.

Presentation. The clear and organized delivery of forensic information by experts in court to make complex evidence understandable and persuasive.

Reliability. The consistency and dependability of forensic methods and expert conclusions, ensuring results can be trusted across different tests and courtroom settings.

CHAPTER 2

RESEARCH METHODOLOGY

2.1 Research Design

This Study will use qualitative research with a descriptive phenomenological approach to portray the specific experiences of a forensic expert. Qualitative research is a scientific methodology focused on gaining a deep and holistic understanding of social phenomena, experiences, and behaviors through non-numerical data. Instead of measuring how many or how much, it seeks to explain the "how," "why," "when," "where," and "who" behind human actions and social interactions. This approach explores the meanings, perspectives, and contextual factors that shape the reality experienced by individuals or groups. Methods often include interviews, observations, focus groups, and document analysis, providing rich and detailed insights into complex issues.

Primary data will be collected through semi-structured, in-depth interviews with participants purposively selected for their extensive experience in court testimonies. These interviews will capture participants' detailed stories, reflections, and sentiments regarding their witness experiences, while also exploring the barriers, adversarial work pressures, and the overall emotional burden of their work. Thematic Analysis will then be conducted, using quoted material to provide a vivid, truthful account of participants' challenges. Interviews will be recorded.

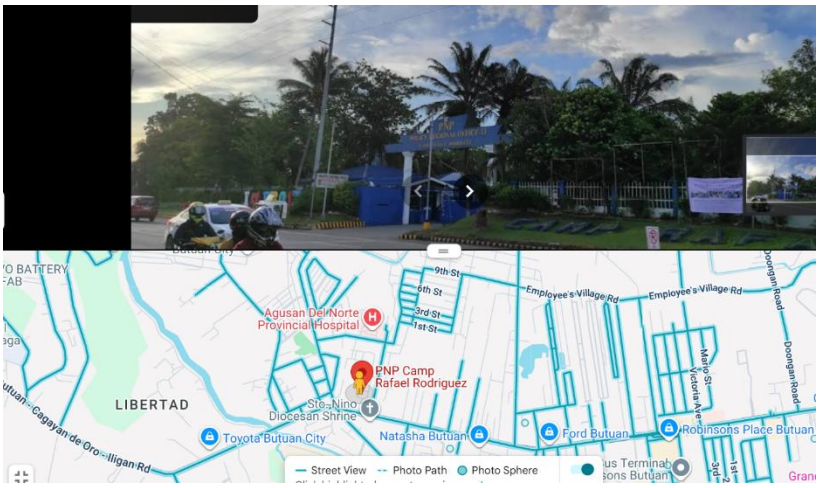
2.2 Research Materials and Methods

This Study employs a qualitative research design to explore the challenges forensic experts face during court presentations. Data were collected through semi-structured interviews with practicing forensic experts with experience testifying in court. The purposive sampling technique was employed to select participants from various forensic disciplines, including biology, chemistry, toxicology, and digital forensics, to ensure a broad perspective on courtroom challenges.

Interview questions focused on communication difficulties, courtroom preparation, use of visual aids, interaction with legal practitioners, and perceptions of juror comprehension. Additional data were collected through a self-developed questionnaire designed to explore forensic experts' personal experiences; it was specifically developed to capture detailed, subjective insights into the obstacles they face in effectively presenting scientific evidence in court. This tool complements observational and document Analysis by providing direct perspectives that deepen the understanding of challenges forensic experts encounter in courtroom settings.

2.3 Research Environment, Population and Sample

This research will focus on the forensic services and expert witnesses at Camp Rafael C. Rodriguez, located in Butuan City, Agusan del Norte, Philippines. The camp, also serving as the seat of command for Police Regional Office 13 (PRO-13), functions as the region-wide central hub for forensic services for the Caraga region and is home to the PRO-13 Crime Laboratory, which employs forensic practitioners in a variety of specialties such as ballistics analysis, fingerprint analysis, and medico-legal forensic services. This site has been selected because it represents the most accessible population of professionals related to the main object of inquiry: forensic and expert witnesses. Additionally, the high concentration of at least 8 to 10 participants in forensic professions at this location will facilitate data aggregation related to the institution and enable an examination of both similarities and differences in the problems faced, as well as their interrelationships among services rendered and related service relations in this particular setting.



2.4 Research Data and Analysis Plan

The researcher will employ a qualitative approach using thematic Analysis to examine the responses of forensic experts and legal practitioners. This method will provide an in-depth understanding of the challenges forensic experts face during courtroom presentations. The Analysis will be guided by the research questions and problem statement, ensuring that key descriptive expressions are carefully identified. Transcribed data was organized into themes to identify common operational obstacles.

The research process involved:

1. Recording: The researchers will document the interviews with forensic experts and stakeholders through audio recording to ensure the accurate capture of responses.

2. Transcription: The recorded interviews will be transcribed into text, ensuring that all details, including tone and emphasis, are preserved to maintain accuracy.

3. Translation: When interviews are conducted in local dialects, the data will be translated into English to ensure consistency and clarity in the Analysis.

4. Coding: The transcribed information will be analyzed for common themes and similar answers. These will then be divided and categorized with codes, forming a basis for structuring the data.

5. Categorization: The information will be categorized into themes to highlight recurring challenges faced by forensic specialists in courtroom testimony, enabling a structured evaluation of the feedback.

The researchers will organize the data by grouping similar ideas into categories and using Atlas. The Ti software will carefully analyze words and themes. This process helps create a well-structured overview of the key topics in the Study.

2.5 Research Ethical Standards

This study adheres to strict ethical standards to safeguard all participants' rights, dignity, and well-being.

A. Protection of Intellectual Property Rights (IPR)

The researchers strictly protected all intellectual property rights associated with forensic methods, expert analyses, and court presentation materials used or referenced in the Study. They recognized the importance of respecting proprietary knowledge and properly crediting original authors, developers, and institutions. The research design included measures to prevent unauthorized use or reproduction of copyrighted or patented materials, thereby facilitating the fair and lawful exchange of scientific information.

B. Informed Consent

They are committed to obtaining informed consent from all participants before participating in the Study. They provide clear, detailed information about the research's objectives, procedures, anticipated risks and benefits, and their rights as participants. This transparency allows participants to make an educated decision about their participation. The researchers also ensure that consent is documented appropriately and that participants can ask questions and receive clarifications.

C. Data Privacy and Confidentiality

The privacy and confidentiality of all data collected were prioritized, especially considering the sensitive nature of forensic work and court presentations. Personal and professional information about the forensic experts was stored securely using encryption, with access restricted to research team members only. To protect identities, data were anonymized or aggregated as necessary. The researchers remained committed to complying with applicable data protection laws and ethical guidelines throughout the research process.

D. Voluntary Participation and Freedom to Withdraw

Participation in this Study was emphasized as entirely voluntary. Individuals were informed that they could choose whether or not to participate. They were assured they

could withdraw from the Study at any point without any negative consequences or loss of benefits. The researchers ensured that participants felt comfortable declining involvement or discontinuing participation if they wished, prioritizing their autonomy and well-being.

E. Minimization of Harm and Risk Management

The researchers acknowledged the potential psychological, professional, or social risks participants might face during the Study. To minimize harm, research activities were carefully designed to be as non-intrusive and low-risk as possible. Support resources were available if needed, and open communication with participants was maintained to address any concerns. Regular risk assessments were also conducted to identify and mitigate unforeseen issues, ensuring participant safety throughout the research.

F. Beneficence and Contribution to Knowledge

The research aimed to generate meaningful insights into the challenges faced by forensic experts during court presentations, to improve professional practices and judicial processes. The researchers intended for the Study's benefits to extend beyond academic circles, contributing to higher-quality forensic testimony and fairer legal outcomes. By advancing scientific knowledge and practical understanding, they fulfilled their ethical responsibility to contribute positively to society.

G. Justice and Fair Participant Selection

Research participants were selected solely based on their relevance to the study objectives, ensuring fairness and impartiality. No discrimination was made based on gender, ethnicity, rank, or other characteristics unrelated to their forensic expertise or role in court presentations. The goal was to achieve a representative sample that reflects the diversity of forensic practitioners, thereby enhancing the generalizability and fairness of the findings.

H. Data Integrity and Accuracy

The researchers upheld rigorous data collection, recording, and Analysis standards to ensure accuracy and reliability. They avoided any fabrication, falsification, or data manipulation and employed multiple validation techniques to ensure the accuracy of their results. Proper documentation and a transparent methodology enabled replication and verification, thereby strengthening the credibility of the research outcomes.

I. Transparency and Honesty in Reporting

The researchers are committed to complete transparency in reporting all findings, including unexpected results, limitations, and potential conflicts of interest. Honest dissemination was upheld to foster trust within the academic community, forensic institutions, and the public. The research outputs were ensured to be accessible and clearly presented, supporting further scholarship and informed decision-making.

J. Use of Patented or Copyrighted Materials

The researchers rigorously checked and secured the necessary permissions or licenses when employing patented tools, software, or copyrighted content. This commitment protected intellectual property rights and supported legal and ethical research practices, preventing infringement and promoting respect for creators' contributions.

K. Ethical Considerations for Animal and Human Rights

The Study focused exclusively on human forensic experts and avoided involving animals. It adhered strictly to human rights standards, ensuring participants' dignity and welfare were respected. There was no coercion, exploitation, or harm inflicted upon participants at any stage. Ethical treatment was fundamental to maintaining the integrity of the research and the forensic profession.

L. Responsible Use of AI and Other Related Technologies

Should the research incorporate artificial intelligence (AI) or other advanced technologies, careful ethical considerations were exercised in their application. The researchers ensured these tools were used responsibly, avoiding biases, inaccuracies, or misuse that could compromise data quality or participants' privacy. Protocols included continuous monitoring and evaluation of technological tools to uphold fairness, transparency, and accountability.

M. Ethical Clearance and Institutional Approval

The research adheres to the highest ethical standards by obtaining formal ethical clearance and institutional approval prior to commencement. This process ensures that the Study respects and safeguards the rights, dignity, and welfare of human forensic expert participants, in alignment with internationally recognized ethical guidelines for research involving human subjects. The research protocol is rigorously reviewed by an accredited Institutional Review Board (IRB) or ethics committee, which assesses the Study's design, methodology, and procedures to minimize risks, prevent harm, and uphold informed consent and confidentiality.

The ethical standard mandates transparent documentation and accountability throughout the research lifecycle to maintain integrity, protect participant privacy, and ensure replicability of findings. Furthermore, any use of advanced technologies, including AI, is continuously monitored to avoid bias or misuse that could compromise data quality or participant rights. Compliance with institutional policies, intellectual property laws, and best practices ensures this Study meets ethical requirements and professional standards essential to trustworthy forensic research and legal credibility.